

Applying NLP for topic modelling in a real-life context

for

Fitness Centres Company

(real company name undisclosed under NDA)

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Contents

PROBLEM STATEMENT.....	3
METHODS	3
EXPERIMENTS AND RESULTS	4
ACTIONABLE INSIGHTS	10
FURTHER ANALYSIS	10

PROBLEM STATEMENT

Fitness Centres operates in a highly competitive market, offering fitness facilities and activities to its members. This analysis focuses on Fitness Centres's operations at its United Kingdom sites, specifically examining member feedback on two customer review platforms: Google and TrustPilot. The analysis will concentrate on negative feedback to identify key customer pain points. Insights from this review will be used to develop strategies and recommendations for Fitness Centres management aimed at improving services and facilities, enhancing customer retention, and strengthening the company's market image. Ultimately, these improvements will support Fitness Centres in consolidating its market position, increasing profitability, and delivering an exceptional gym experience to its members.

METHODS

The insights from customer reviews are derived with use of the following methods:

Method	Key Model Features	Method Outline
Natural Language Processing (NLP) BERTopic	- The name is an acronym of bi-directional encoder representation from transformers - simultaneously processes text from left and right context - powerful in Natural Language Processing (NLP) Embedding: use of vectors Clustering: based on dimensionality reduction similar themes are grouped into one topic Offers a high degree of logical and semantic understanding however it is sufficiently refined to understand language nuance e.g. sarcasm	Topics: Identifies main topics (latent themes) in provided text based on statistical probabilities: Text converted into vector embeddings (using transformer BERT) Clusters: groups similar text into clusters Key words and probabilities: identified inside each cluster based on most frequently occurring words. Visualisation: allows graphical representation of topics to analyse their clustering via intertopic distance map, dendrograms, and keyword bars.
Large Language Models (LLMs) phi4-mini	Capable of understanding nuances in language (e.g. sarcasm) and coherent text generation Capable of reasoning Suitable for prompting / dialogue	Does not automatically output topic list (like BERTopic) but it can be prompted to do so
Latent Dirichlet Allocation (LDA) Gensim	Uncovers latent data patterns with use of probabilistic methods	Uses topic and word distributions which are updated based on observation in recurrent fashion Uses Dirichlet distribution to vector parametrise continuous multivariate probability distributions

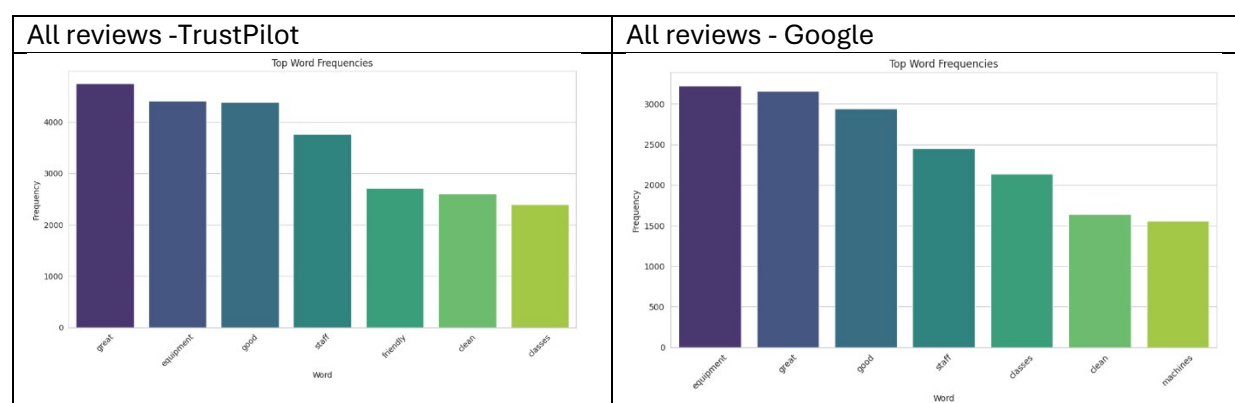
DATA STRUCTURE

Data is provided in two csv files containing reviews collected from Google and TrustPilot sites.

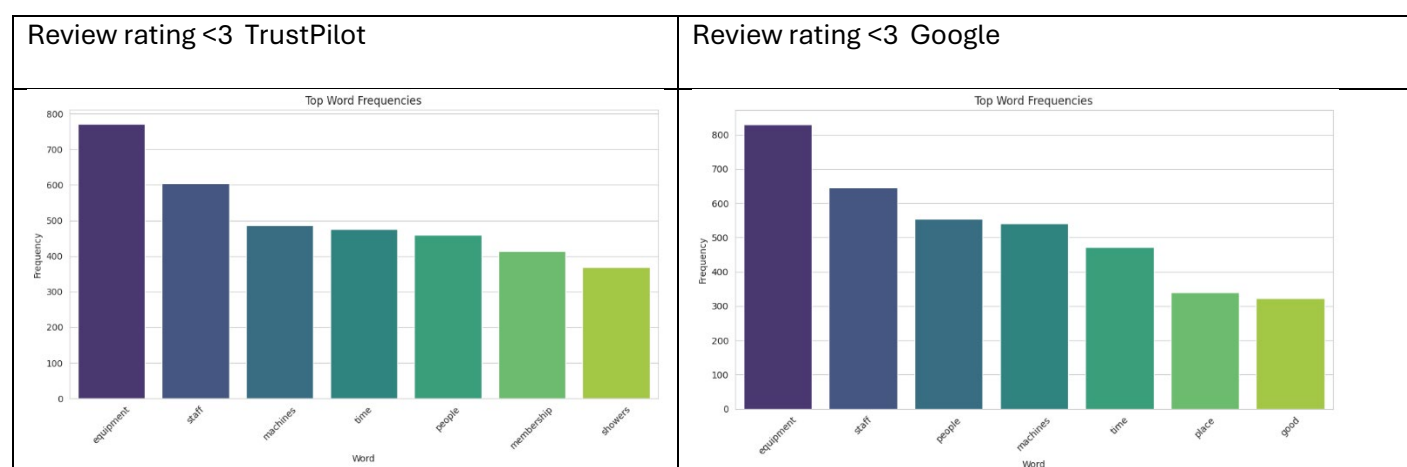
The data is cleaned and standardised as follows across both data sets to the following categories essential for further analysis: Creation Date, Overall Score, Club's Name, Comment.

EXPERIMENTS AND RESULTS

Initial analysis of individual datasets did not reveal any specific issues. Positive words such as *great*, *good*, and *clean* were the most frequently occurring terms, appearing in positions 1, 3, and 6 in TrustPilot reviews, and 2, 3, and 5 in Google Reviews. While this may provide a superficial sense of reassurance, it risks overlooking reviews from dissatisfied customers potentially impacting customer retention and the Fitness Centres brand.



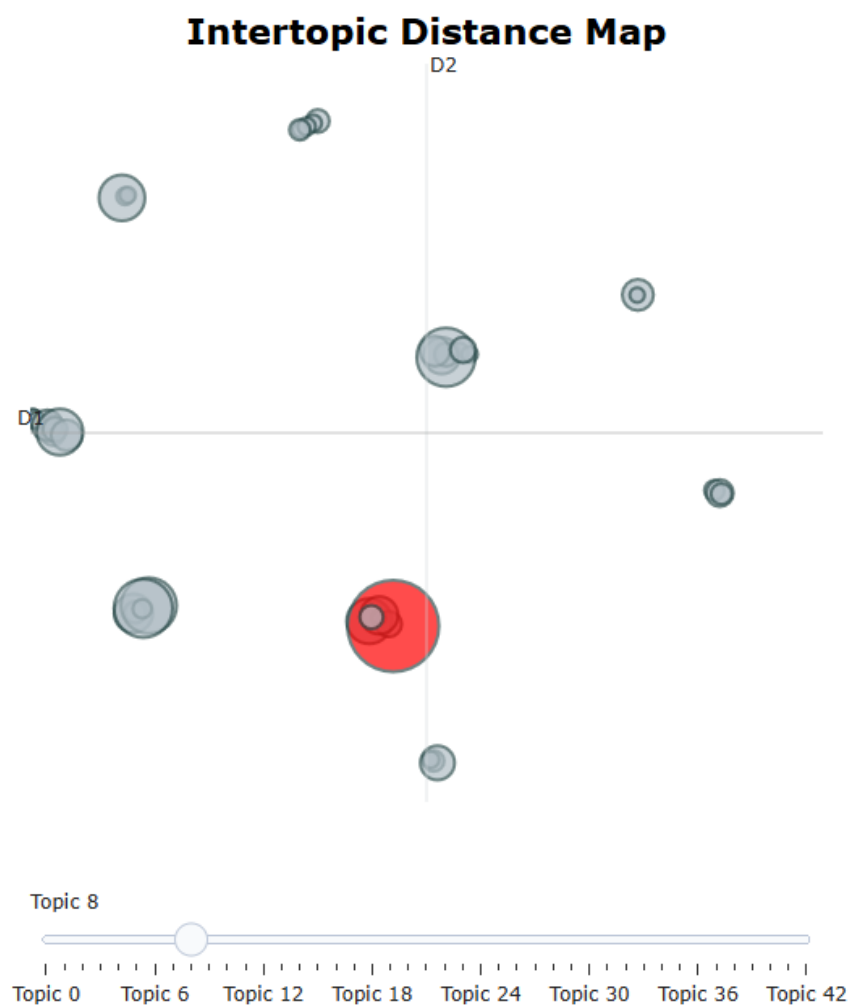
Focusing on negative reviews (ratings below 3), several issues emerge from the dataset analysis. The previously most frequently occurring word, *equipment*, now appears in a negative context and is the top term in both datasets' negative reviews. Similarly, *machines* ranks among the most frequent negative words. Other recurring topics in negative reviews across both datasets include *staff*, *time*, and *people* (referring to interactions with other members).



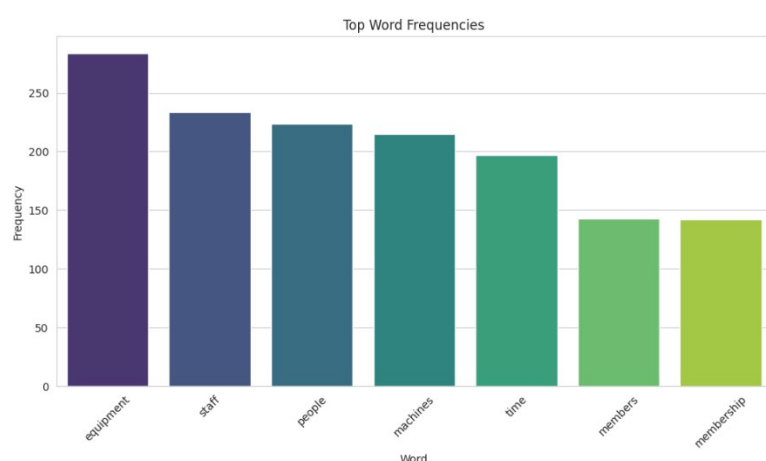
Analysis of all negative reviews using BERTopic models revealed latent themes, providing more granular insights into customer dissatisfaction. The main areas of concern identified are:

1. Confusing layout of the gym
2. Inefficient air conditioning, particularly during the summer
3. Management and staff perceived as unfriendly or lacking a member-focused attitude
4. Insufficient space in some classes or cancellations of scheduled classes
5. Substandard hygiene and cleanliness in changing areas and toilets
6. Problems with gym equipment
7. Parking issues and unclear information on parking and parking tickets
8. Loud music volume, potentially too high for some members
9. Unclear opening hours during holidays, particularly around Christmas
10. Poor cleanliness of showers

These topics form distinct clusters, demonstrating that BERTopic effectively identified unique and actionable customer pain points.



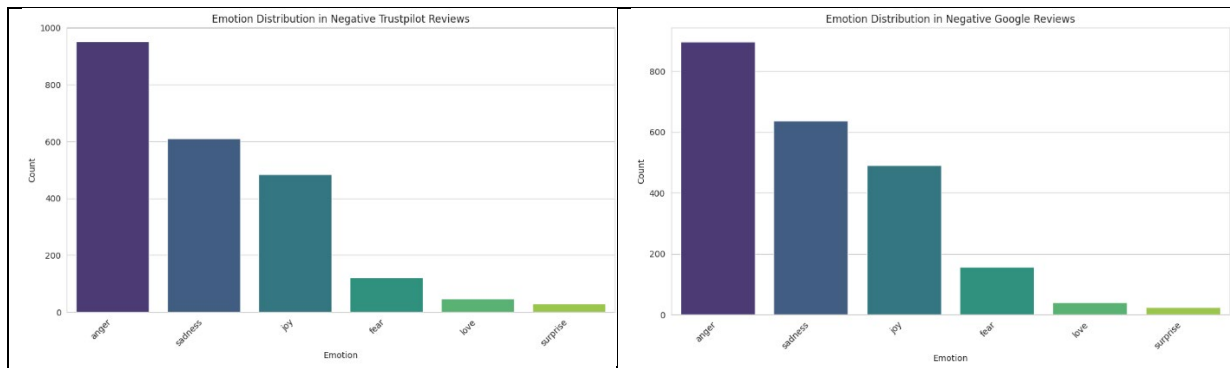
Analysis of the top 20 locations with the highest number of negative reviews on TrustPilot and Google shows that eight locations overlap between both datasets, six of which are in London. Among the remaining locations, London consistently features as having the highest concentration of negative reviews. Word analysis of the top 30 locations indicates that many issues are likely driven by high population density, resulting in more customers per gym, heavier use of equipment, and greater strain on facilities such as changing rooms and showers. Additionally, reduced customer-focused attention contributes to dissatisfaction. Correspondingly, complaints most frequently mention *equipment* and *machines*, followed by *staff* and interactions with other members (*people*). Some issues related to membership also emerge.



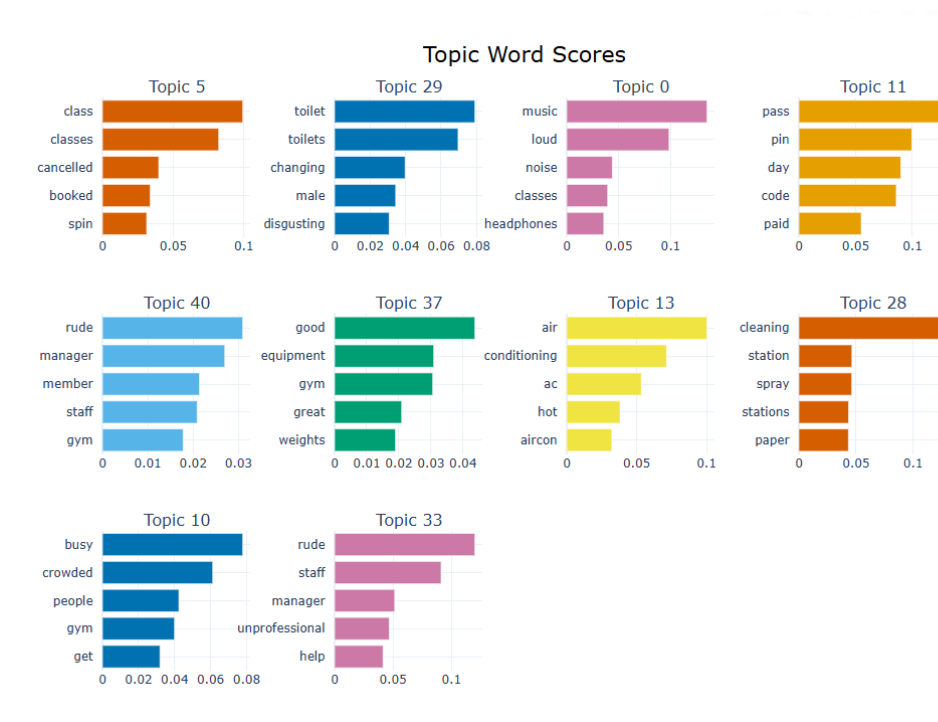
Fine-tuning these findings with BERTopic compresses customer negative reviews into key themes. **Parking** (Topic 1) emerges as a primary concern, followed by **gym atmosphere and usage** (Topic 2), encompassing issues with equipment, staff, and interactions with other members. Additional insights are obtained through **emotions analysis**, providing further understanding of customer sentiment and intensity of dissatisfaction.

TrustPilot Reviews		Google Reviews	
joy	7884	joy	7660
anger	1581	anger	1537
sadness	1063	sadness	1027
love	352	love	331
fear	296	fear	307
surprise	114	surprise	78

This aligns with earlier word analysis, where *great* and *good* were among the most frequently occurring words. Focusing on reviews with ratings below 3, **anger** is the dominant emotion (953 reviews on TrustPilot and 898 on Google), followed by **sadness**. Notably, even reviews with ratings above 3—which are not classified as negative—often express **anger** (an additional 628 reviews on TrustPilot and 639 on Google), indicating that even satisfied customers experience specific pain points.

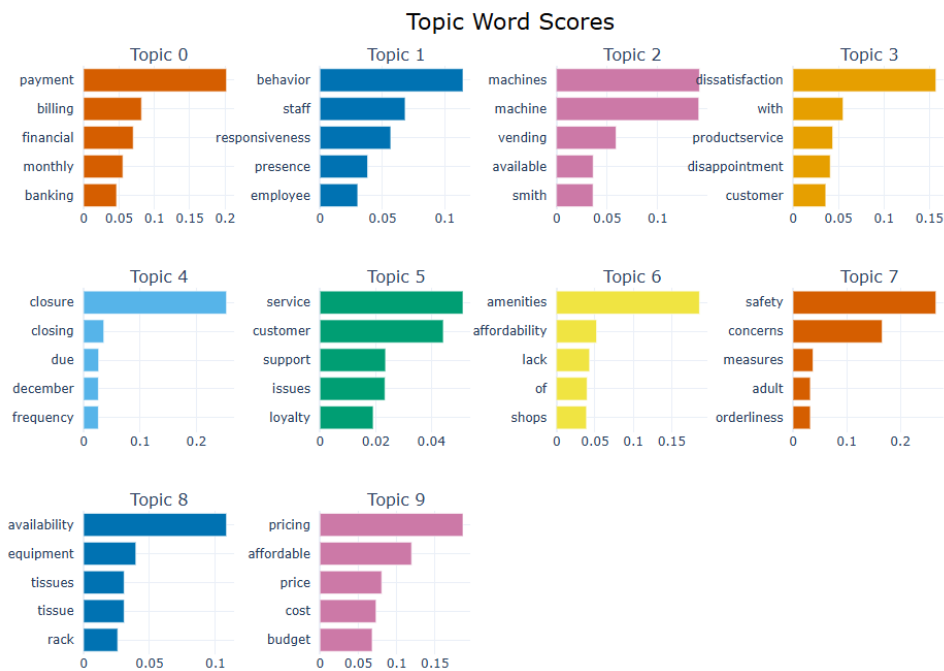


Analysis of the **top 10 topics from angry reviews** largely mirrors the themes identified in all negative reviews but provides additional granularity into the sources of customer anger. Key topics include **class overbooking or cancellations, poor hygiene in toilets and changing facilities, loud music, issues with pin codes, rude or unprofessional staff and management, and inefficient air-conditioning**. These insights highlight the specific operational and service-related triggers driving strong negative emotions among gym users.

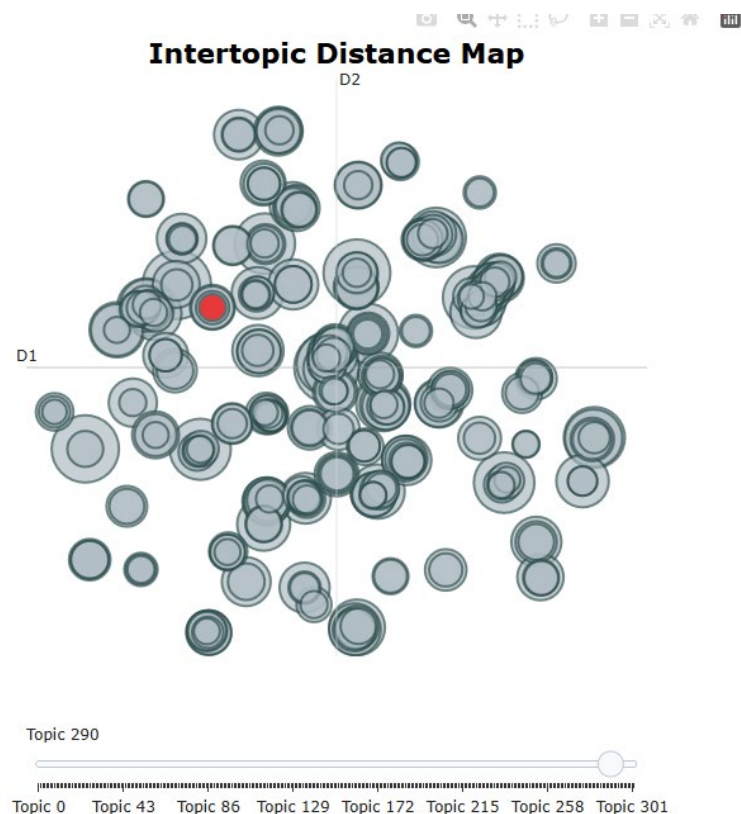


Large Language Models enable more sophisticated analysis of customer feedback (see Word bar chart below). Topic analysis using **phi4-mini with BERT** uncovered new, previously unreported issues, including **monthly billing problems** (84 reviews), more detailed **staff-related concerns** (lack of responsiveness or insufficient presence on the gym floor), and **equipment issues**, including vending machines (74 reviews). Other emerging themes include **overall customer dissatisfaction** (73 reviews), **gym closures in December**, **substandard customer support**, **safety concerns**, and **high pricing combined with low cost transparency** and lack of amenities.

These findings provide a deeper understanding of operational and service pain points impacting customer loyalty.

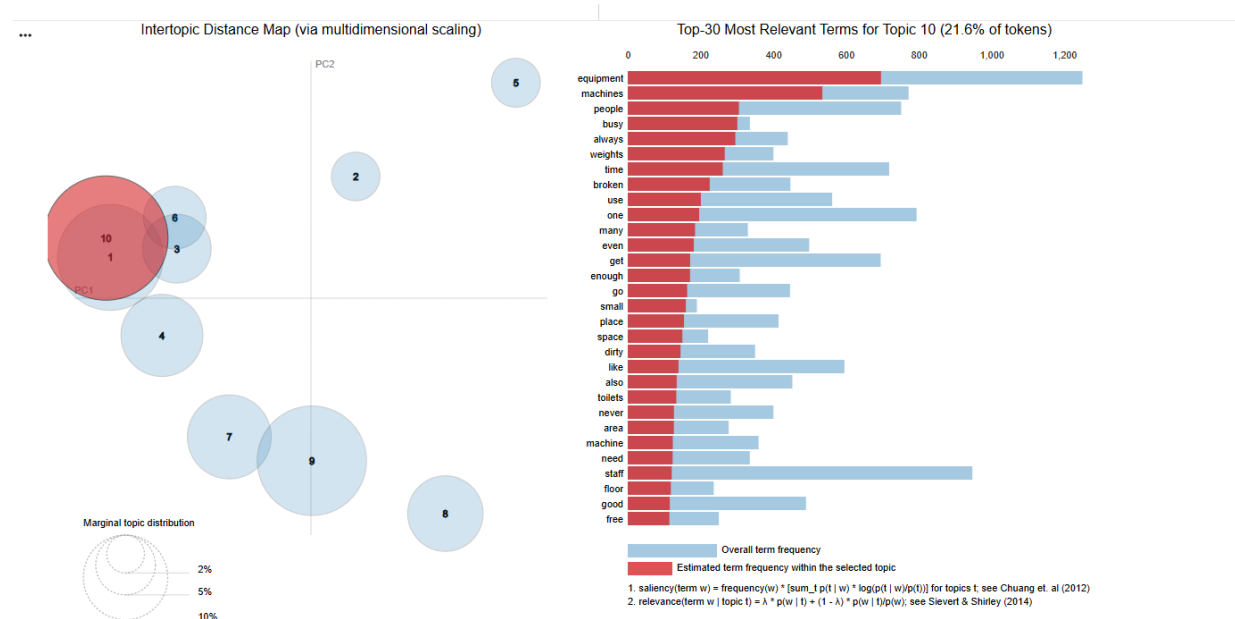


A large number of themes were identified in customers' negative feedback, enabling more detailed analysis and the design of targeted remediation plans.



Genism (LDA) analysis results were consistent with the themes identified by BERT. The most salient topic in negative reviews was **equipment and machines** (visualised below), often described as broken, dirty, or overcrowded—particularly weights—followed by **maintenance issues** such as air conditioning, cleanliness of facilities (gym floor, showers, bathrooms), and **staff-related concerns** including lack of courtesy or helpfulness and poor service. Other notable topics included **parking problems**, **gym atmosphere** (busy spaces and other members), and **difficulties booking classes**.

While BERT analysis demonstrated higher coherence, LDA offered a more granular breakdown of themes. BERT’s topics, in contrast, had better compression, reducing the likelihood that the same problem was split across multiple themes. An additional benefit of LDA is the ability to quantify the **marginal contribution of each word to a specific topic**, alongside overall word frequency, providing a nuanced view of term importance within topics.



In summary, this analysis leveraged multiple data slices and methodologies—**NLP (BERT)**, **LLM (phi4-mini)**, and **Gensim (LDA)**—which converged on broadly similar customer pain points. These revolve around:

- Quality of interactions with gym staff
- Quality, availability, and accessibility of equipment
- Premises accessibility, including parking
- Overall cleanliness
- Courtesy of other gym members
- Gym experience affected by peak-time congestion and class/equipment availability

BERT performed particularly well on LLM outputs, producing clearly refined topics with high topic dispersion while also uncovering new insights, such as issues with **monthly billing** and **membership plans/value for money**.

ACTIONABLE INSIGHTS

Continuing analysis with **phi4-mini** highlighted recurring issues, which can be addressed with the following measures:

1. **Facilities:** Improve maintenance, focusing on hygiene in bathrooms and gym areas.
2. **Customer Experience:** Provide staff with customer service training, optimize staff-to-member ratios, and offer member engagement events.
3. **Equipment:** Ensure all equipment is functional, maintain availability of cardio machines, and improve the layout of the weights area.
4. **Operations:** Communicate opening hours clearly, maintain consistent water temperature in showers, and resolve air-conditioning issues, particularly during summer.
5. **Payments & Memberships:** Ensure timely, transparent payments with no unexpected charges, and review pricing to ensure perceived value for money.
6. **Accessibility:** Review and improve parking options and ensure accessibility for all members.

It is recommended that **Fitness Centres management implement these remediation plans** to enhance customer satisfaction, improve retention, and strengthen brand image.

FURTHER ANALYSIS

Further analysis can be commissioned to provide deeper insights into customer pain points by location, with particular focus on London Fitness Centres sites, which ranked in the top 20 for negative feedback across both TrustPilot and Google datasets.